A Comprehensive Reliability Assessment of hBN-encapsulated MoS_2 FETs and Inverters

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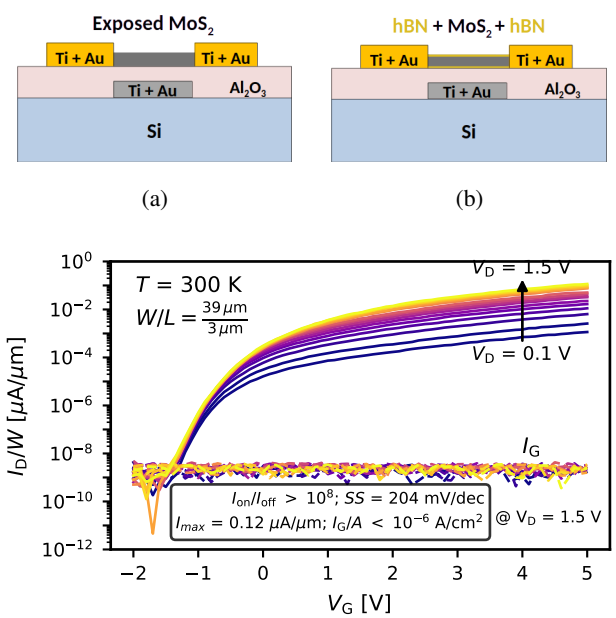
I. Introduction

II. Methods and Fabrication



(a) (b)

(c)

Fig. 1. **Cross-section schematics**: (a) exposed and (b) hBN-encapsulated MoS_2 FETs. (c) **Performance of the fabricated devices**: $I_{D(G)}$-V_G curves for different drain voltages and main device metrics.

Acknowledgment

The preferred spelling of the word "acknowledgment" in America is without an "e" after the "g". Avoid the stilted expression "one of us (R. B. G.) thanks ...". Instead, try "R. B. G. thanks...". Put sponsor acknowledgments in the unnumbered footnote on the first page.

We gratefully acknowledge funding from the European Innovation Council under project number 101223249.

References

Please number citations consecutively within brackets [1].

References

[1] J. F. Fuller, E. F. Fuchs, and K. J. Roesler. "Influence of harmonics on power distribution system protection". In: *IEEE Trans. Power Delivery* 3 (Apr. 1988), pp. 549–557.

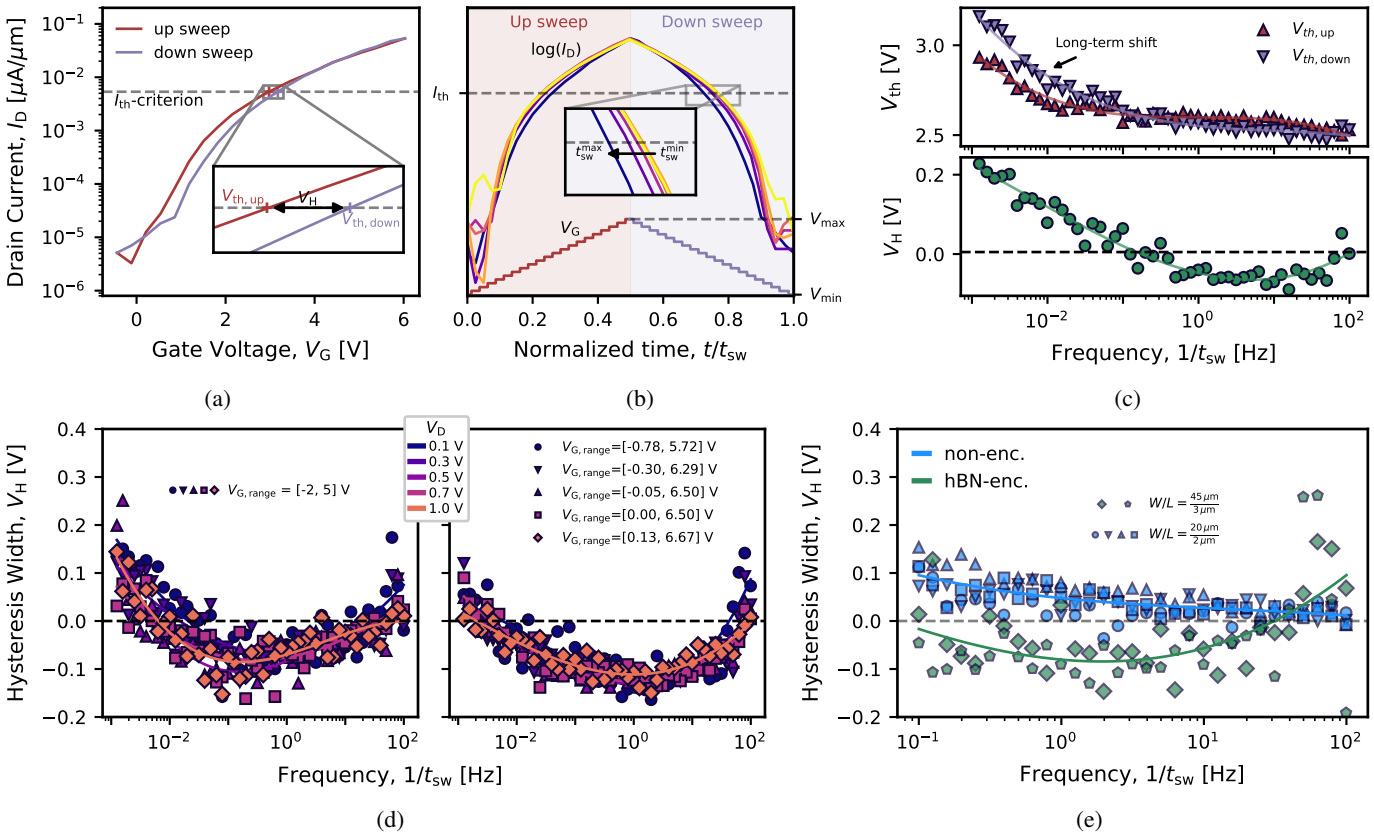


Fig. 2. (a) **Typical hysteresis curve**: the hysteresis width, V_{H}, is defined as the difference between the threshold voltages extracted during the up and down sweeps, i.e., $V_H = V_{\mathrm{th,up}} - V_{\mathrm{th,down}}$. (b) **Hysteresis dependence on the gate voltage sweep rate**: $\log(I_D)$ and V_{G} are shown as a function of the normalized sweep time, t/t_{sw}, for different sweep rates, r_{sw}. (c) **Hysteresis-r_{sw} curves**: Threshold voltages, $V_{\mathrm{th,up}}$ and $V_{\mathrm{th,down}}$, and hysteresis width, V_{H}, are shown as a function of the sweep rate, r_{sw}.

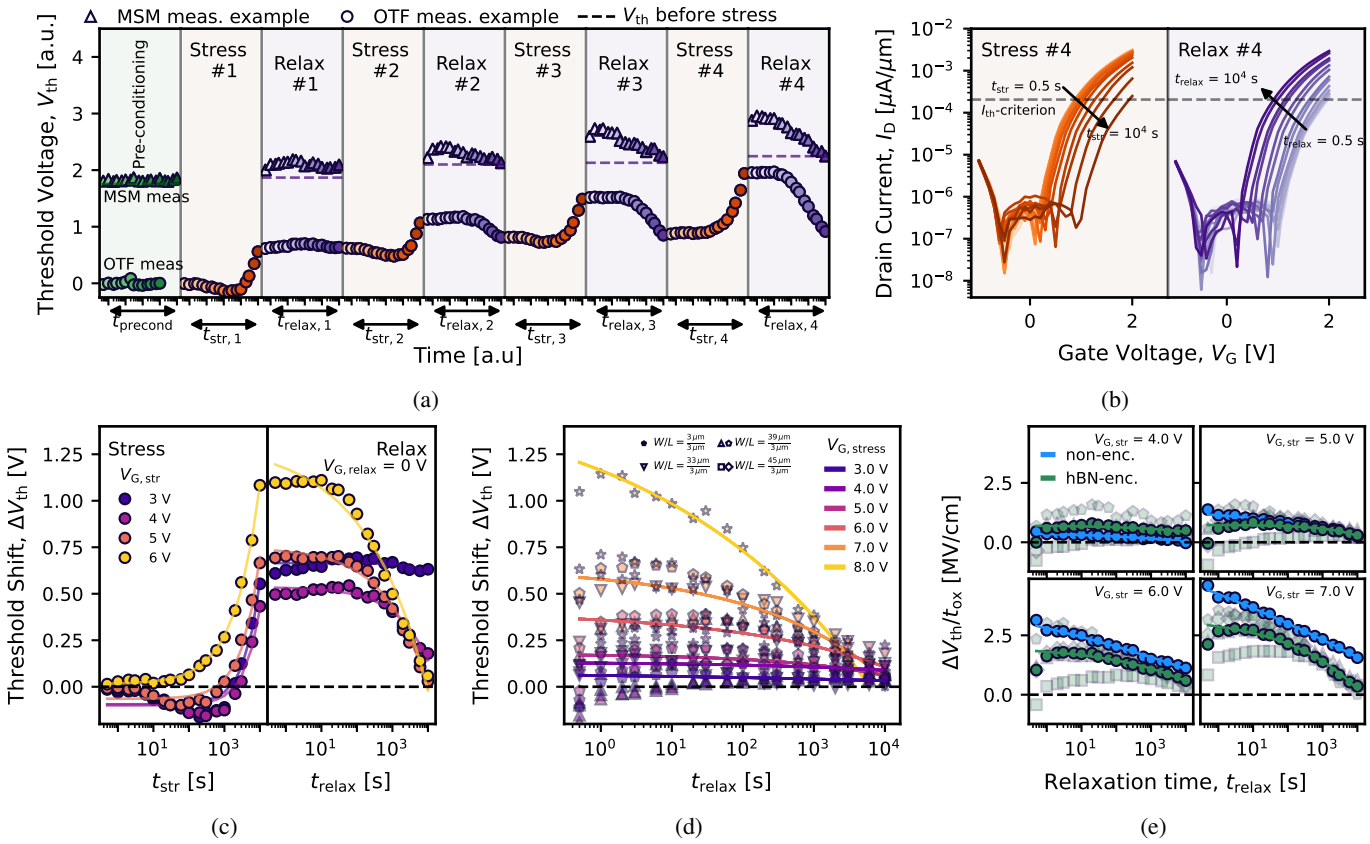


Fig. 3. (a) Example of a hysteresis curve and the corresponding time evolution of the gate voltage. (b) Comparison of the $I_{ds} - V_{gs}$ curves for the RO stage and the asymmetric stage.

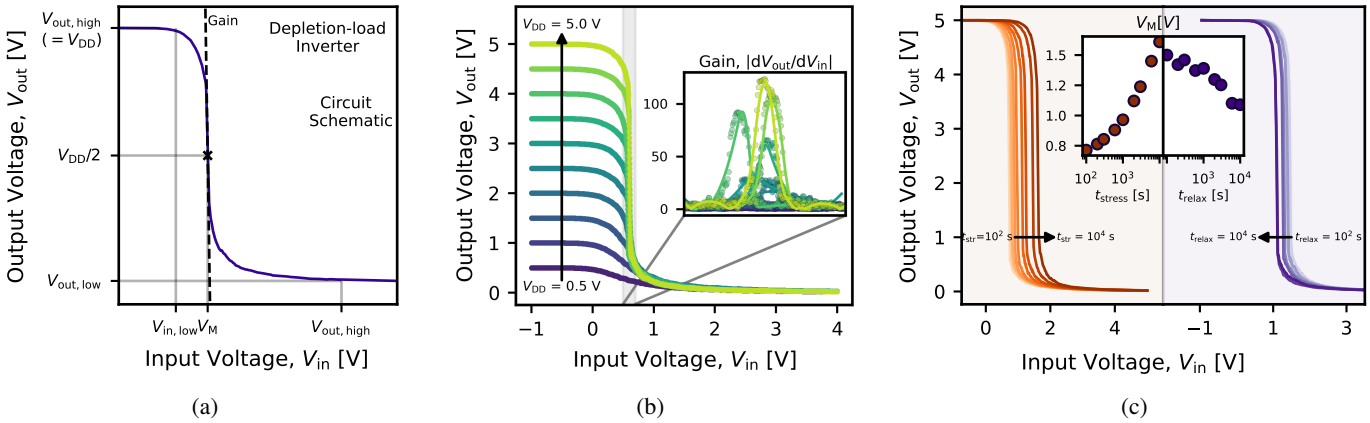


Fig. 4. (a) Example of a TDDB curve and the corresponding time evolution of the gate voltage. (b) Comparison of the $I_{ds} - V_{gs}$ curves for the RO stage and the asymmetric stage.